Γ-Net: From the First Law of Thermodynamics to Three Inviolable Constraints

Framework Definition and Theorem Directory

Hsi-Yu Huang[1,*]

[1] *Independent Researcher, Taipei, Taiwan*

(Dated: March 22, 2026)

We derive the complete axiomatic framework of the Γ-Net theory from a single physical origin: the First Law of Thermodynamics applied to one-dimensional interfaces. Three constraints emerge as *theorems*, not postulates: C1 (energy conservation: $\Gamma^2 + T = 1$), C2 (coefficient update: $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$), and C3 (dimensional relay: K-staircase necessity). C2 is proven to be the *unique* first-order activity-gated update rule that minimises reflected energy. We provide a complete inventory of derived quantities and a theorem directory mapping the deductive chain across five companion papers: Paper 1 (existence and irreducibility), Paper 2 (geometric quantification and metabolic scaling), Paper 3 (temporal unification and remodeling equivalence), Paper 4 (topological pathology), Paper 5 (full-system verification). This paper contains no new results beyond the derivation itself; it serves as the master reference for the series.

I. INTRODUCTION

The Γ-Net framework describes biological organisation as impedance physics. Every interface between two biological media—synapse, blood vessel wall, bone–cartilage junction, cell membrane, skin surface—is characterised by an impedance mismatch that determines how much energy is transmitted versus reflected.

The framework rests on three constraints, C1–C3, which are not modelling choices but consequences of the First Law of Thermodynamics. This paper derives all three from first principles and proves the uniqueness of the central remodeling law (C2).

Five companion papers develop the consequences:

- **Paper 1** [1]: Proves that any finite network *must* have irreducible topological cost ($A_{\rm cut} > 0$), derives the brain as the unique convergence structure, and extends to memory, consciousness, and nociception as impedance dynamics.
- **Paper 2** [2]: Quantifies the geometric cost: derives Murray's law, Kleiber's 3/4-power metabolic scaling, dual-network organ health, and the interface scale table spanning 14 orders of magnitude.
- **Paper 3** [3]: Proves C2 is algebraically equivalent to all 29 known tissue-remodeling laws, derives impedance debt, sleep, the lifecycle equation, aging, and emotion as temporal consequences of a single remodeling rule.
- **Paper 4** [4]: Derives disease as topological perturbation: E0 emergence of cardiovascular, neural, metabolic, and coupled multi-organ pathology from C1/C2/C3 alone, with zero disease-specific code.
- **Paper 5** [5]: Validates the full framework against NHANES epidemiological data (n = 49,774, zero fitted parameters), reports all-cause mortality AUC = 0.705, and collects all falsifiable predictions from Papers 1–4.

A reader entering at any single paper encounters local axiom recaps but lacks: (i) the derivation chain from the First Law to C1/C2/C3; (ii) the proof of C2 uniqueness; (iii) a unified quantity inventory; (iv) a theorem directory mapping every major result to its location in the series. This paper fills those four gaps.

II. Γ AS THERMODYNAMIC NECESSITY

A. Impedance at a one-dimensional interface

Consider any one-dimensional interface separating two media with generalised impedances Z_1 and Z_2, where impedance is defined as the ratio of effort (voltage, pressure, force, temperature) to flow (current, volume flow, velocity, heat flux):

$$Z_k = \frac{\text{effort}_k}{\text{flow}_k}\,. \tag{1}$$

Theorem 1 (Γ universality)**.** *At any one-dimensional interface between media of impedance Z_1 and Z_2, the First Law of Thermodynamics requires that incident power be partitioned into reflected and transmitted fractions:*

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}\,, \qquad T = 1 - \Gamma^2 = \frac{4Z_1Z_2}{(Z_1+Z_2)^2}\,, \tag{2}$$

where Γ is the reflection coefficient and T the transmission coefficient.

Proof. At a lossless one-dimensional interface, conservation of energy (First Law) requires: $P_{\rm incident} = P_{\rm reflected} +$

* llc.y.huangll@gmail.com

$P_{\text{transmitted}}$. The boundary conditions on effort and flow (continuity of effort, conservation of flow) uniquely determine the reflected fraction as $|\Gamma|^2 = |(Z_2 - Z_1)/(Z_2 + Z_1)|^2$. This derivation applies identically to electrical transmission lines, acoustic waveguides, hydraulic conduits, mechanical interfaces, and thermal boundaries—the mathematical structure depends only on the existence of effort and flow variables, not on the physical medium. □

Theorem 1 is not a modelling choice. Any system with two impedances meeting at an interface *must* obey this partition. The reflection coefficient Γ is as fundamental as temperature or pressure: it exists wherever two impedances meet.

B. The Minimum Reflection Principle

For an organism with N interface channels, the total reflected (wasted) power is:

$$P_{\text{waste}} = \sum_{i=1}^{N} \Gamma_i^2 \, P_{\text{in},i} \, . \tag{3}$$

An organism is *thermodynamically viable* if and only if

$$P_{\text{metabolic}} > P_{\text{waste}} \, , \tag{4}$$

i.e., its metabolic power production exceeds its reflection losses. Organisms that violate Eq. (4) die by energy deficit—no natural-selection argument is required.

The Minimum Reflection Principle (MRP) is therefore not an optimality *assumption* but a thermodynamic *viability condition*: organisms that waste too much energy at their interfaces fail by physics alone.

Define the **reflection action** over lifetime T:

$$\boxed{\mathcal{A}[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t) \, P_{\text{in},i}(t) \, dt \;\rightarrow\; \min \, .} \tag{5}$$

This is the variational principle from which all three constraints are derived.

III. THE THREE CONSTRAINTS

A. C1: Energy Conservation

Theorem 2 (C1)**.** *At every interface i, at every instant t:*

$$\Gamma_i^2(t) + T_i(t) = 1 \, , \tag{6}$$

where $T_i = 1 - \Gamma_i^2$ is the transmission coefficient.

Proof. By Theorem 1, at any lossless one-dimensional interface the reflected power fraction is $|\Gamma_i|^2$ and the transmitted fraction is $T_i = 4Z_1Z_2/(Z_1 + Z_2)^2$. Direct substitution of $\Gamma_i = (Z_2 - Z_1)/(Z_2 + Z_1)$ yields:

$$\Gamma_i^2 + T_i = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} + \frac{4Z_1Z_2}{(Z_2 + Z_1)^2} = \frac{(Z_2 - Z_1)^2 + 4Z_1Z_2}{(Z_2 + Z_1)^2} = \frac{(Z_2 + Z}{(Z_2 + Z} \tag{7}$$

Dissipative corrections modify $Z_i(t)$ over time (Johnson–Nyquist noise, Arrhenius aging) but at each instant the power balance at the interface is governed by the instantaneous impedances, so C1 holds pointwise in time. □

C1 is the First Law applied locally. It holds exactly for any lossless interface. Dissipative corrections (Johnson–Nyquist noise, Arrhenius aging) modify Z over time but never violate power balance at a single interface.

Remark 1 (Signal power vs. metabolic work)**.** *C1 governs instantaneous* signal power *at each interface: incident = reflected + transmitted. The metabolic energy fuelling C2 update enters as a separate thermodynamic work term that modifies Z itself, not the $\Gamma^2 + T = 1$ identity. The analogy is an electronic amplifier: the DC power supply changes the gain (modifies Z), yet Kirchhoff's power law holds at every RF port.*

Corollary 1 (Thermal Dissipation)**.** *The reflected fraction Γ_i^2 of incident power is re-absorbed by the source impedance and dissipated as Joule heat:*

$$P_{\text{heat}} = \sum_{i=1}^{N} \Gamma_i^2 \, P_{\text{in},i} \, . \tag{8}$$

The body temperature in thermal steady state satisfies $P_{\text{heat}} = hA\,(T_{\text{body}} - T_{\text{env}})$. The impedance mismatches that perform computation therefore generate the heat that sustains the temperature at which those computations remain viable.

B. C2: Coefficient Update

Theorem 3 (C2 as gradient descent on $\mathcal{A}$)**.** *First-order gradient descent on the reflection action $\mathcal{A}$ (Eq. 5) with respect to Z_i, gated by input and output activity, yields:*

$$\boxed{\Delta Z_i = -\eta \, \Gamma_i \, x_{in,i} \, x_{out,i} \, ,} \tag{9}$$

where $\eta > 0$ is the update rate, $x_{in,i}$ is the input signal, and $x_{out,i}$ is the transmitted signal.

Proof. The functional derivative of $\mathcal{A}$ with respect to Z_2 at an interface gives:

$$\frac{\delta \mathcal{A}}{\delta Z_2} = \frac{4Z_1(Z_2 - Z_1)}{(Z_2 + Z_1)^3} \, P_{\text{in}} \propto \Gamma \, . \tag{10}$$

Gradient descent $\Delta Z = -\eta \, (\delta \mathcal{A}/\delta Z)$ with activity gating (update only when both input and output are active) yields Eq. (9). □

C2 is *not a postulate*: it is the unique first-order activity-gated update rule that minimises reflected energy (Section IV). Its form is identical to Hebb's learning rule when Z is synaptic conductance, to Wolff's law when Z is bone stiffness, to Glagov's remodeling when Z is vascular compliance, and to 26 further tissue-specific laws (Paper 3 [3]).

Historical note. In the original formulation, C2 was called the "Hebbian update" with signals x_{pre}, x_{post}. Paper 3 proves that the same equation governs all 29 tissue types, making neural-specific terminology misleading. We adopt *coefficient update* and $(x_{\text{in}}, x_{\text{out}})$ throughout.

C. C3: Dimensional Relay

Theorem 4 (C3 (E0))**.** *If two nodes i, j have modal complexities $K_i \neq K_j$, then $\Gamma_{ij} = 1$ (total reflection). C2 cannot remove it because there is no Z-mismatch to minimise—the receiving dimension simply does not exist. The only resolution is to insert intermediate relay nodes forming a K-staircase $K_i > K_{m_1} > K_{m_2} > \cdots > K_j$, splitting the single $(K_i - K_j)$-mode total reflection into smaller reflections, each of which C2 can partially reduce.*

Proof. A mode in i with no corresponding degree of freedom in j sees infinite load impedance $\to \Gamma = 1$. This is topological, not impedance-based: C2 can only adjust Z within existing dimensions. By induction, a chain of relays $K_i > K_{m_1} > \cdots > K_j$ minimises total Γ^2 subject to the topological constraint. □ □

C3 explains why biological systems develop layered architecture:

- *Neural*: cortex ($K = 5$) → thalamus → brainstem → spinal cord → motor neuron → muscle ($K = 1$).
- *Vascular*: aorta → arteries → arterioles → capillaries—a fractal K-staircase for impedance matching.
- *Skeletal*: bone → cartilage → tendon → muscle—graded stiffness across the joint interface.

Without dimensional relay, high-K controllers and low-K effectors would be separated by 100% reflection walls, and complex life would be physically impossible.

Remark 2 (Metadata co-transport)**.** *Every energy or material flow across an interface carries (Z_{source}, Z_{load}) as metadata, so that Γ_i is locally computable without global network knowledge. This is not an additional physical law but a necessary consequence of Theorem 1: since Γ_i depends only on the local pair $(Z_{source,i}, Z_{load,i})$, each carrier (electrical signal, blood, mechanical load) must encode its source impedance upon departure.*

Implementation note. In the reference implementation (ALICE), C3 is enforced by requiring all inter-module values to be `ElectricalSignal` objects carrying Z metadata. Raw floating-point numbers between modules are forbidden.

IV. MATHEMATICAL UNIQUENESS OF C2

Theorem 5 (C2 uniqueness)**.** *Among all first-order, local, activity-gated update rules of the form $\Delta Z = f(Z, x_{in}, x_{out})$, the rule $\Delta Z = -\eta\, \Gamma\, x_{in}\, x_{out}$ is the* unique *descent direction that minimises $\mathcal{A}[\Gamma]$ at every step.*

Proof. Any first-order descent must follow $\Delta Z \propto -\delta\mathcal{A}/\delta Z$. By Eq. (10), $\delta\mathcal{A}/\delta Z \propto \Gamma$. Activity gating restricts updates to epochs where both $x_{\text{in}} \neq 0$ and $x_{\text{out}} \neq 0$. The unique factorisation satisfying locality, activity gating, and descent is $\Delta Z = -\eta\, \Gamma\, x_{\text{in}}\, x_{\text{out}}$ with $\eta > 0$. □

A. The tissue is not the algorithm

The molecular machineries implementing C2 differ vastly across tissues: NMDA receptors (neural), RANKL/OPG (bone), eNOS (vascular), AID somatic hypermutation (immune), keratinocyte Ca^{2+} (skin), HGF (liver), EPO (marrow), FSH receptors (reproductive)—completely different molecular pathways implementing the same equation. The substrate determines the material constants (η, Z_0); the equation is dictated by physics.

B. Objective function for viability

The Γ-framework supplies a *physical* objective function: $\mathcal{A} = \int_0^T \sum_i \Gamma_i^2\, P_{\text{in},i}\, dt$. Any system whose $\mathcal{A}$ exceeds the viability bound is physically excluded. This is a constraint on what *can* exist, not an explanation of what *does* exist.

C. Sex as a boundary condition

Sex does not alter C1–C3. The hypothalamic–pituitary–gonadal (HPG) axis sets sex-specific Z_{setpoint} values: testosterone raises η_{muscle}; oestrogen modulates Z_{adipose} and Z_{bone} (menopause $\equiv$ $Z_{\text{bone}} \uparrow$, $\Gamma_{\text{bone}} \uparrow$); pre-menopausal oestrogen maintains lower Γ_{vasc} via enhanced endothelial C2. Sex is a parametric shift in boundary conditions, not a different dynamical law.

V. DERIVED QUANTITIES

Table I lists every named quantity in the Γ-Net framework, ordered by the paper where it first appears.

VI. THEOREM DIRECTORY

The five companion papers form a deductive chain radiating outward from C1/C2/C3. Each paper occupies a unique mathematical role. Table II provides the complete map.

A. The deductive chain

The logical structure is a concentric expansion:

Layer	Content
0	First Law → Γ, C1, MRP, C2, C3 (this paper)
1	C2 on finite network → $A_{\text{cut}} > 0$ → topology, brain,
2	C2 on branching geometry → Murray, Kleiber, H
3	C2 ≡ 29 laws → D_Z, sleep, lifecycle, aging (Paper
4	Topology + perturbation → disease emergence (E0)
5	Γ-vector + NHANES → validation, blueprints, predi

Each layer uses only C1/C2/C3 plus the results of inner layers. No layer imports concepts from an outer layer. This unidirectional dependency is the structural guarantee against circularity.

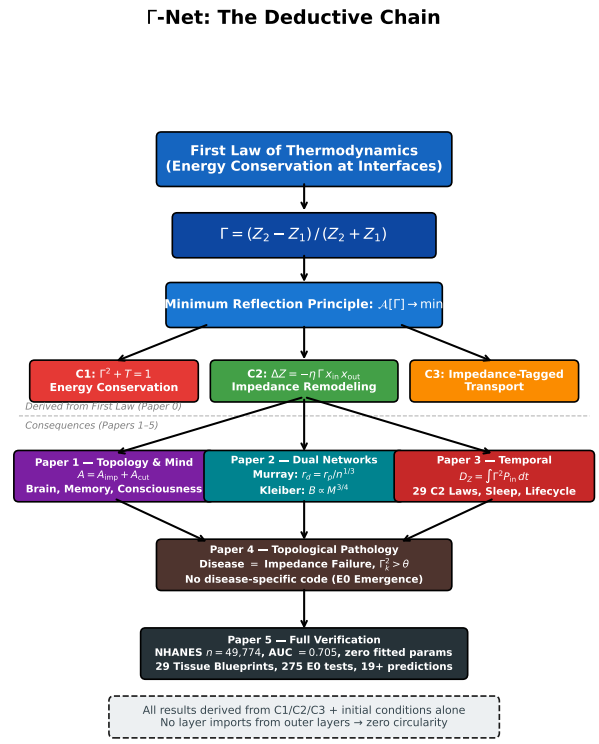

FIG. 1. The deductive chain of the Γ-Net framework. From the First Law of Thermodynamics, Γ emerges as thermodynamic necessity; the Minimum Reflection Principle yields C1/C2/C3; each subsequent paper applies these constraints to a specific domain. No layer imports from an outer layer.

VII. WHAT THIS THEORY DOES NOT CLAIM

The Γ-framework derives *physical constraints* that any living system must satisfy. It does *not* reconstruct thermodynamic history.

1. **Necessary conditions, not sufficient causes.** The framework shows that a K=5 brain *must* maintain $T > T_c(5)$ and *must* have vasculature. It does not explain *why* such brains appeared or *which* lineages developed them.

2. **No historical trajectory.** Statements such as "the tidal zone offers the lowest-action path" identify geometric optima in impedance space. Whether any lineage actually traversed that path is a question for palaeontology, not for this framework.

3. **C2 uniqueness is mathematical, not biological.** The proof that C2 is the unique first-order $\mathcal{A}$-minimising rule is a theorem about gradient descent. That multiple molecular machineries converge on the same equation is *consistent with* C2 being a physical attractor; it is not evidence that history "discovered" C2.

4. **Compatibility, not replacement.** Thermodynamics determines which lineages survive; the Γ-framework supplies the physical objective function that selection operates on.

Throughout the six papers, any sentence referencing the relationship to biology should be read as *physical constraint language*: "X is required" means "any system with property Y must satisfy X".

VIII. NOTATION CONVENTIONS

Table III collects notation used across Papers 0–5.

IX. CONCLUSION

The Γ-Net framework is derived, not postulated. The First Law of Thermodynamics at a one-dimensional interface yields the reflection coefficient Γ. Energy conservation gives C1 ($\Gamma^2 + T = 1$). Gradient descent on the

TABLE I. Complete quantity inventory of the Γ-Net framework.

Symbol	Name	Definition	Interpretation	Paper
Γ_i	Reflection coefficient	$\frac{Z_{\text{load},i}-Z_{\text{source},i}}{Z_{\text{load},i}+Z_{\text{source},i}}$	Mismatch at interface i	0
T_i	Transmission coefficient	$1-\Gamma_i^2$	Fraction of power delivered	0
$\mathcal{A}$	Reflection action	$\int_0^T \sum_i \Gamma_i^2 P_{\text{in},i}\, dt$	Total wasted energy	0
A_{imp}	Improvable action	$\mathcal{A} - A_{\text{cut}}$	Reducible by C2 update	1
A_{cut}	Irreducible action	$\sum_{k>K^*} \Gamma_k^2$	Topological cost immune to C2	1
Φ_{meta}	Consciousness index	$1-\Gamma_{\text{meta}}^2$	Self-referential matching	1
$\mathcal{S}$	Null space ("soul")	$\ker(\partial Z/\partial t)$	Impedances immune to experience	1
β	Murray bifurcation ratio	$2^{1/3} \approx 1.26$	Optimal branch radius ratio	2
Ω	Dual-network overhead	≥ 1	Packing cost of two trees	2
H	Health index	$\prod_i (1-\Gamma_i^2)$	Whole-body transmission	2
D_Z	Impedance debt	$\int_0^{T_{\text{wake}}} \lvert\Gamma\rvert^2 P_{\text{in}}\, dt$	Cumulative mismatch; cleared by sleep	3
Lifecycle	Lifecycle ODE	$\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\,\Sigma\Gamma^2 + \gamma\,\Gamma_{\text{env}} + \delta\, D(t)$	Cognition bathtub curve	3
Π_Γ	Gamma-Pain Index	$w_n\Gamma_n^2 + w_v\Gamma_v^2 + w_D \frac{dD_Z}{dt}$	Consciousness-independent pain	4

reflection action gives C2 ($\Delta Z = -\eta\,\Gamma\, x_{\text{in}}\, x_{\text{out}}$), proven unique among first-order activity-gated rules. Local computability gives C3.

From these three constraints, five companion papers derive, in a strictly outward deductive chain: topology and the mind (Paper 1), geometric quantification and metabolic scaling (Paper 2), temporal unification across 29 tissues (Paper 3), disease as topological pathology (Paper 4), and full-system verification against epidemiological data (Paper 5).

This paper provides the master reference: a derivation chain (Sections II–IV), a quantity inventory (Table I), and a theorem directory (Table II).

All source code, simulation scripts, and NHANES pipelines are available at `https://github.com/cyhuang76/alice-gamma-net` under AGPL-3.0.

[1] H.-Y. Huang, "The irreducibility theorem and the physics of mind: topology, memory, consciousness, and nociception from impedance constraints," companion Paper 1 (2026).

[2] H.-Y. Huang, "Geometric quantification: Murray's law, Kleiber scaling, and dual-network organ health from Γ-minimisation," companion Paper 2 (2026).

[3] H.-Y. Huang, "Temporal unification: C2 as twenty-nine tissue laws, impedance debt, sleep, and the lifecycle equation," companion Paper 3 (2026).

[4] H.-Y. Huang, "Topological pathology: disease emergence from C1/C2/C3 without disease-specific code," companion Paper 4 (2026).

[5] H.-Y. Huang, "Full-system verification: NHANES validation, tissue blueprints, and falsifiable predictions," companion Paper 5 (2026).

[6] H.-Y. Huang, *ALICE: A physics-driven electronic lifeform*, `https://github.com/cyhuang76/alice-gamma-net` (2026).

TABLE II. Theorem directory: the deductive chain from C1/C2/C3 to all major results. Each row is a theorem or major result, with its paper of origin and the constraints required for its derivation. Theorems are grouped by paper and by mathematical role.

Paper	**Role**	**Theorem / Result**	**Requires**	**Section**
Paper 1: Existence and Irreducibility				
	Existence	$\mathcal{A} = A_{\text{imp}} + A_{\text{cut}}$, $A_{\text{cut}} > 0$	C1, C2, K-dim cascade	P1 §2–3
	Topology	Same-K clustering, relay hierarchy	$A_{\text{cut}} > 0$, C2	P1 §4–5
	Convergence	Brain as unique convergence structure	Relay hierarchy	P1 §6
	Memory	Memory $\equiv dZ/dt$ (three regimes)	C2	P1 §8
	Consciousness	$\Phi_{\text{meta}} = 1 - \Gamma^2_{\text{meta}}$; Television Theorem	C1, C2, feedback	P1 §9–10
	Soul	$\mathcal{S} = \ker(\partial Z/\partial t)$; four laws of null space	C2 uniqueness	P1 §11
	Nociception	Three-layer pain; Π_Γ index	C1, C2, SNR	P1 §13–14
Paper 2: Geometric Quantification				
	Vascular	Murray's law from Γ-minimisation; $\beta = 2^{1/3}$	C2, Poiseuille Z	P2 §2–4
	Scaling	$A_{\text{cut}} \propto M^{2/3}$; Kleiber $B \propto M^{3/4}$	A_{cut}, Murray, Ω	P2 §7–8
	Dual-network	$H = (1-\Gamma_n^2)(1-\Gamma_v^2)$; four failure modes	C1, C2, two buses	P2 §5–6
	Universality	Interface scale table (14 orders of magnitude)	Γ universality (Thm 1)	P2 §9
Paper 3: Temporal Unification				
	Equivalence	C2 $\equiv$ 29 tissue-update laws	C2, tissue substitutions	P3 §2
	Sleep	Sleep $= D_Z$ discharge; gate-leak inequality	C2, D_Z, Johnson–Nyquist	P3 §3–5
	Lifecycle	$\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\Sigma\Gamma^2 + \gamma\Gamma_{\text{env}} + \delta D$	C2, D_Z, Arrhenius	P3 §6
	Aging	Coffin–Manson fatigue; death as $\Gamma^2_{\text{basal}} \geq 1$	C2, null space, D_Z	P3 §7
	Emotion	Dopamine = RPE; Yerkes–Dodson; curiosity	C2, D_Z	P3 §8
	Morphogenesis	Dual-field PDE ($\partial Z/\partial t$, $\partial\rho/\partial t$)	C2, reaction–diffusion	P3 §9
Paper 4: Topological Pathology				
	E0 cardio	HTN, MI, aortic stenosis from topology	C1/C2/C3, GammaTopology	P4 §2
	E0 neural	ALS, stroke, PTSD, phantom limb from topology	C1/C2/C3, GammaTopology	P4 §3
	Coupled	Five-organ cascade; three recovery regimes	C2, coupling matrix	P4 §4–5
	Metabolic	T2DM six-channel hexagonal deadlock	C2, bifurcation	P4 §6
	Necrosis	Tissue necrosis as substrate collapse	$D_Z > D^{\max}_{\text{repair}}$	P4 §7
	Sleep path.	Insomnia / narcolepsy / hypersomnia	Gate-leak inequality	P4 §8
Paper 5: Full-System Verification				
	NHANES	7/7 organs AUC significant; $C = 0.70$; 6.89× risk ratio	Γ universality, H	P5 §2
	Propagation	Cross-organ AUC gain (cardiac +0.166)	Network coupling	P5 §3
	Blueprints	29 tissue blueprints; 278 nodes, 426 edges	C1/C2/C3, K-cascade	P5 §4
	Ablation	C2-on/C2-off pincer tests	E0 emergence	P5 §5
	Predictions	19+ falsifiable predictions (unified list)	Papers 1–4	P5 §6

TABLE III. Notation conventions used across all six papers.

Symbol	**Meaning**	**Units / Range**	**Paper**
Z	Impedance	Ω etc.	0
Γ	Reflection coefficient	$[-1, +1]$	0
T	Transmission $(1-\Gamma^2)$	$[0, 1]$	0
η	Update rate	> 0	0
$(x_{\text{in}}, x_{\text{out}})$	Interface signals	signal units	0
$\mathcal{A}$	Reflection action	J·s	0
K	Modal complexity	integer ≥ 1	1
A_{imp}	Improvable action	J·s	1
A_{cut}	Cutoff action	J·s	1
Φ_{meta}	Consciousness index	$[0, 1]$	1
$\mathcal{S}$	Null space	—	1
β	Murray ratio $(2^{1/3})$	dimensionless	2
Ω	Dual-network overhead	≥ 1	2
H	Health index	$[0, 1]$	2
Z_v, Z_n	Vascular / neural Z	domain-specific	2
D_Z	Impedance debt	J·s	3
$T_c(K)$	Critical temperature	°C	3
Π_Γ	Γ-Pain Index	dimensionless	4